$$= \sum_{j=0}^{N} b_j x(n-j) - \sum_{j=1}^{N} a_j y(n-j)$$

PROCEDURE

1. Open MATLAB
2. Open new M-file
3. Type the program
4. Save in current directory
5. Compile and Run the program
6. For the output see command window/Figure window

$y(-1) = 1$

$y(n) = x(n) + y(n-1)$

THEORETICAL CALCULATIONS

n	x(n) = sin(2*pi*f*n)	y(n)
0	sin(2*pi*5*0) = 0	y(0) = 0 + 1 ⇒ y(n) = 1
1	sin(2*pi*5*1) = 0.5213	y(1) = 0.5213 + 1 = 1.5213
2	sin(2*pi*5*2) = 0.8897	y(2) = 0.8897 + 1.5213 = 2.411
3	sin(2*pi*5*3) = 0.9973	y(3) = 0.9973 + 2.411 = 3.4083
4.	sin(2*pi*5*4) = 0.81245	y(4) = 0.81245 + 3.4083 = 4.22075
5.	sin(2*pi*5*5) = 0.38946	y(5) = 0.38946 + 4.22075 = 4.61021
6.	sin(2*pi*5*6) = -0.14774	y(6) = -0.14774 + 4.61021 = 4.46247

MATLAB PROGRAM

```
% sinusoidal waveform generation using
  recursive difference equation.
clc;
close all;
clear all;
I = input ('enter initial conditions : y(-1)');
n = 0:0.01:1
f = input ('enter the freq');
x = sin(2 * pi * f * n)
y = zeros (1, length (n));
for i = 1 : length (n);
    y(i) = x(i) + I;
    I = y(i);
end
Subplot (411);
plot (n,x);
xlabel ('time (seconds)');
ylabel ('amp');
title ('sinusoidal waveform using recursive
        difference equation');
Subplot (412);
plot (n,y);
xlabel ('time (seconds)');
ylabel ('amplitude');
title ('sinusoidal waveform using recursive
        difference equation');
```

o/p:-

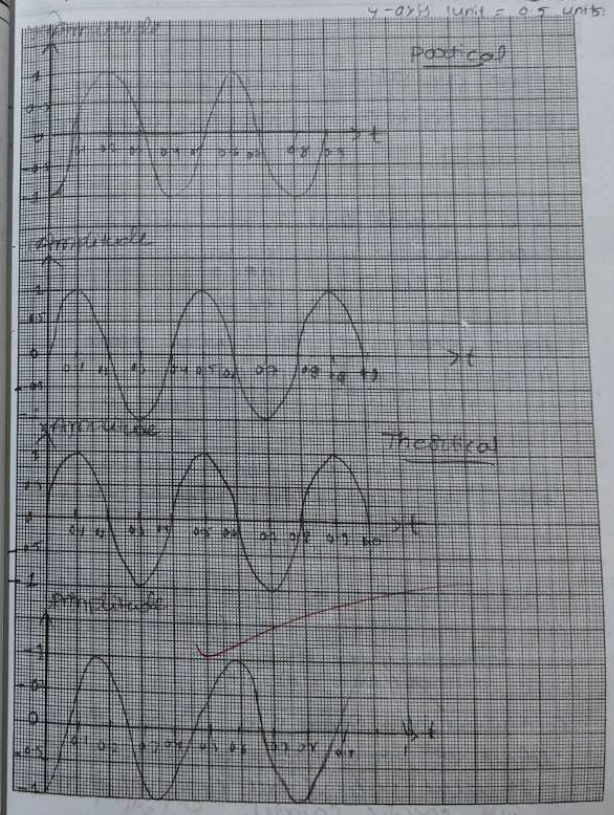
enter initial conditions: y(-1): 1

enter the freq 5.

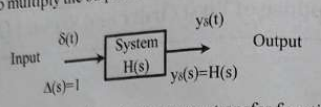
Scale:-

x-axis 1unit = 0.01 units

y-axis 1unit = 0.5 units



In this diagram the input is the unit step function, $\gamma(t)$, and the output is the unit step response, $y_\gamma(t)$. If we multiply the input in Laplace by "s" (i.e., we differentiate the input function in time), we also multiply the output by "s" (or differentiate the step output).



The impulse response of the system is given by the system transfer function. For this reason the impulse response is often called h(t).

PROCEDURE

1. Open MATLAB
2. Open new M-file
3. Type the program
4. Save in current directory
5. Compile and Run the program
6. For the output see command window/Figure window

THEORETICAL CALCULATIONS

let us consider first order difference equation

$y(n) - 2y(n-1) = x(n) + 2x(n-1)$

Apply z-transform on both sides

$Y(z) - 2z^{-1}Y(z) = X(z) + 2z^{-1}X(z)$

$Y(z)[1-2z^{-1}] = X(z)[1+2z^{-1}]$

$H(z) = \frac{Y(z)}{X(z)} = \frac{1+2z^{-1}}{1-2z^{-1}}$

$Y(z) = -1 + \frac{2}{1-2z^{-1}}$

$$\begin{array}{r|l} & -1 \\ -2z^{-1}+1 & 2z^{-1}+1 \\ & -2z^{-1}+1 \\ \hline & 2 \end{array}$$

Apply inverse z-transform

$y(n) = -\delta(n) + 2(2)^n u(n)$

Step

Step as input

$\frac{Y(z)}{X(z)} = \frac{1+2z^{-1}}{1-2z^{-1}}$ $[\because X(z) = 1]$ → Step. $\left[\because 1 = \frac{z}{z-1}\right]$

$Y(z) = \frac{z}{z-1} \cdot \frac{1+2z^{-1}}{1-2z^{-1}}$

$\frac{Y(z)}{z} = \frac{1}{z-1} \times \frac{1+2z^{-1}}{1-2z^{-1}} = \frac{1+2z^{-1}}{(1-z^{-1})(1-2z^{-1})}$

$\frac{A}{1-z^{-1}} + \frac{B}{1-2z^{-1}} = \frac{1+2z^{-1}}{(1-z^{-1})(1-2z^{-1})}$

$1+2z^{-1} = A(1-2z^{-1}) + B(1-z^{-1})$

put $z^{-1} = 1$

$3 = -1A \Rightarrow \boxed{A = -3}$

put $z^{-1} = \frac{1}{2}$

$2 = \frac{1}{2}B \Rightarrow \boxed{B = 4}$

$Y(z) = \frac{-3z}{1-z^{-1}} + \frac{4z}{1-2z^{-1}}$

Apply inverse z-transform

$y(n) = -3(1)^n u(n) + 4(2)^n u(n).$

OUTPUT

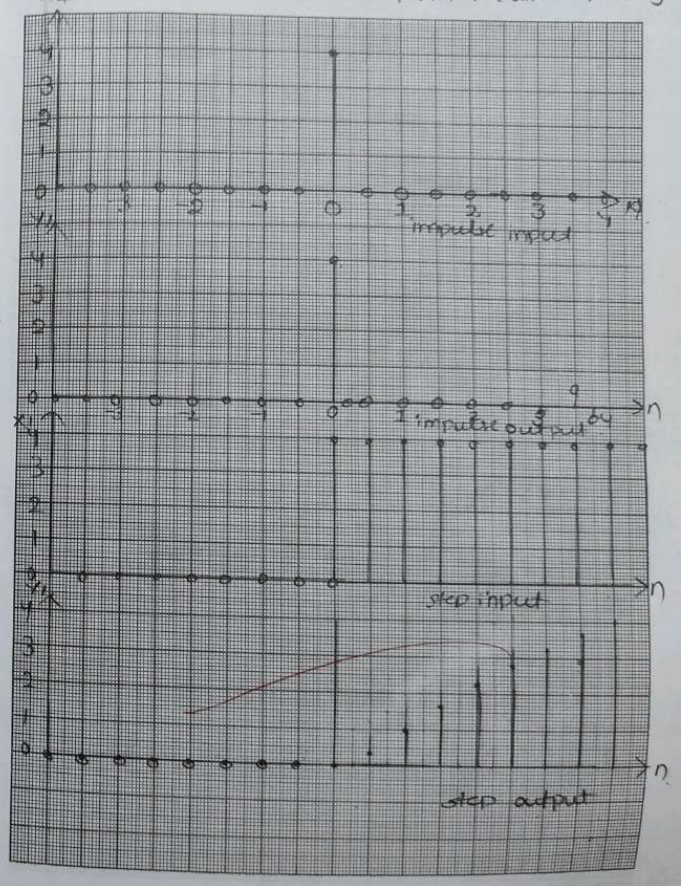
for impulse

```
clc;
clear all;
close all;
n= -10:1:10;
b= input('the coefficient of x(n) =');
a= input('the coefficient of y(n) =');
x1 = 1 * (n==0);
y1 = filter(b, a, x1); (% automatical solving the
subplot (2,1,1);          different equation
stem (n, x1);             -> filter).
title('impulse input plot');
subplot (2,1,2);
stem (n, y1);
xlabel ('n');
ylabel ('y1');
title ('impulse output');
```

for step:

```
clc;
clear all;
close all;
b= input('the coefficient of x(n)=');
a = input('the coefficient of y(n)=');
x1 = 1 * (n>=0);
y1 = filter (b, a, x1);
subplot (2,1,1);
stem (n, x1);
xlabel ('n');
ylabel ('x1');
title ('step input');
subplot (2,1,2);
stem(n, y1);
xlabel ('n');
ylabel('y1');
title('step output');
```


x-axis : 1 unit = 0.5 units
y-axis : 1 unit = 1 unit



5. Compile and Run the program
6. For the output see command window/Figure window

THEORETICAL CALCULATIONS

Given difference eqn is $y(n) = x(n) + 0.9\,x(n-2) - 0.4\,y(n-2)$

taking fourier transform on both sides

$$Y(e^{j\omega}) = X(e^{j\omega}) + 0.9\,e^{-j2\omega}X(e^{j\omega}) - 0.4\,e^{-j2\omega}Y(e^{j\omega})$$

$$Y(e^{j\omega})\left[1+0.4e^{-j2\omega}\right] = X(e^{j\omega})\left[1+0.9e^{-j2\omega}\right]$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1+0.9e^{-j2\omega}}{1+0.4e^{-j2\omega}}.$$

$$H(e^{j\omega}) = \frac{1+0.9\cos 2\omega - j0.9\sin 2\omega}{1+0.4\cos 2\omega - j0.4\sin 2\omega}.$$

magnitude response $|H(e^{j\omega})| = \dfrac{\sqrt{(1+0.9\cos 2\omega)^2+(0.9\sin 2\omega)^2}}{\sqrt{(1+0.4\cos 2\omega)^2+(0.4\sin 2\omega)^2}}$

$$= \frac{\sqrt{1+1.8\cos 2\omega + 0.81}}{\sqrt{1+0.8\cos 2\omega + 0.16}}$$

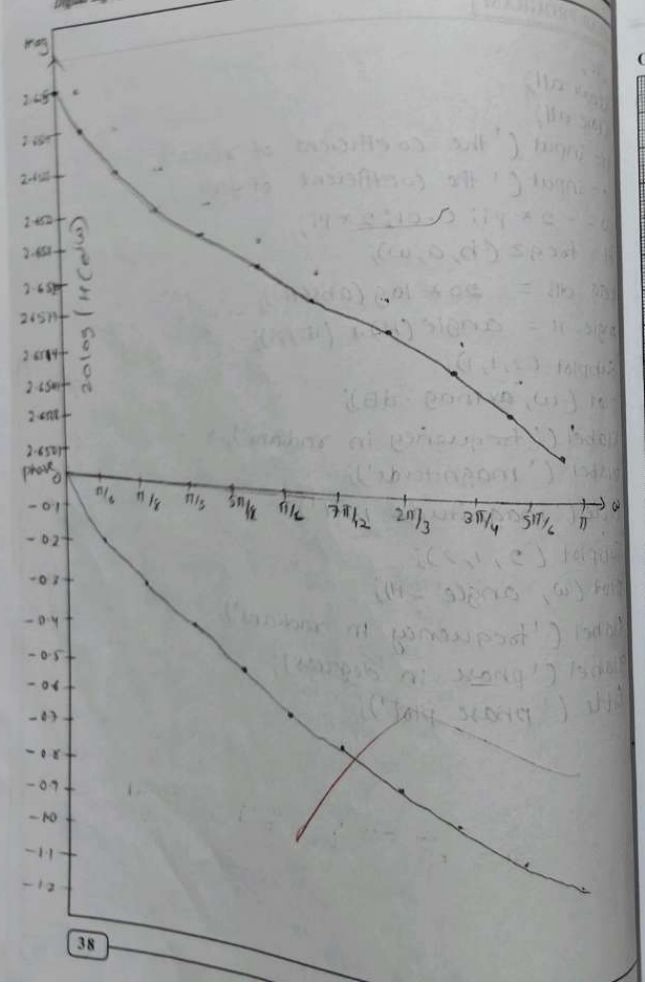
$$= \frac{\sqrt{1.81+1.8\cos 2\omega}}{\sqrt{1.16+0.8\cos 2\omega}}$$

phase response

$$\angle H(e^{j\omega}) = \tan^{-1}\left(\frac{-0.9\sin 2\omega}{1+0.9\cos 2\omega}\right) - \tan^{-1}\left(\frac{-0.4\sin 2\omega}{1+0.4\cos 2\omega}\right)$$

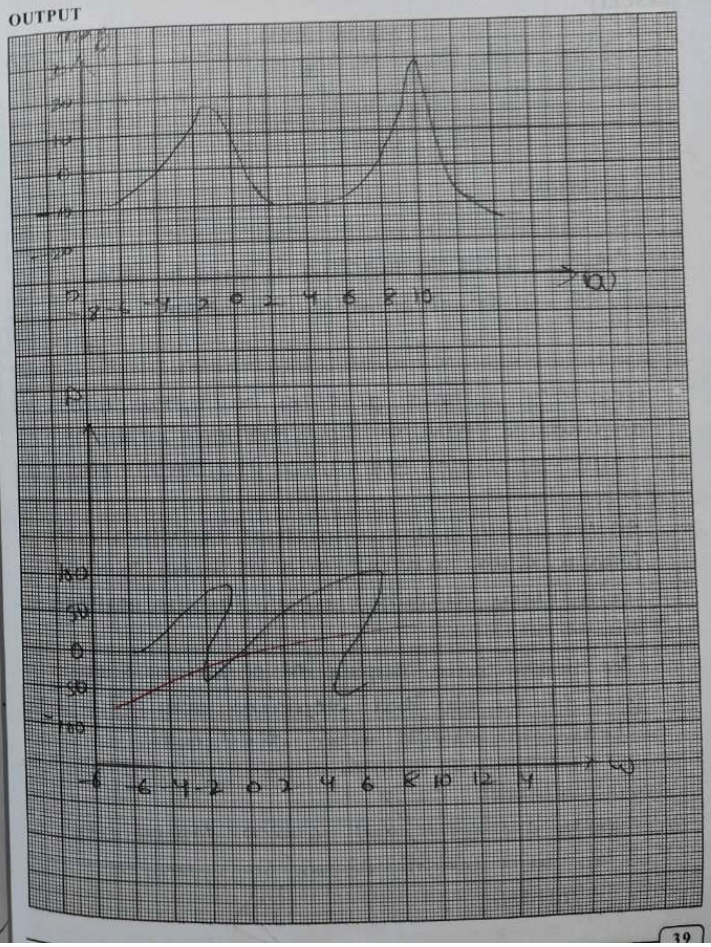
MATLAB PROGRAM

```
clc;
clear all;
close all;
b= input ('the coefficient of x(n) =');
a = input (' the coefficient of y(n) =');
ω = - 2 * pi: 0.01: 2 * pi;
H = freqz (b, a, ω);
mag - dB = 20 * log (abs(H));
angle - H = angle (H) * (180/pi);
subplot (2,1,1);
plot ( ω, mag - dB);
xlabel (' frequency in radians');
ylabel (' magnitude');
title (' magnitude plot');
subplot (2, 1, 2);
plot (ω, angle - H);
xlabel ('frequency in radians');
ylabel ('phase in degrees');
title (' phase plot');
```



scale :-
x-axis : 1 unit = 1 cm
y-axis : 1 unit = 20 cm.

OUTPUT



spectrum, k=1 corresponds to 1kHz frequency. For the same fs and x[n], if N=80 point DFT is computed, then in the resulting spectrum, k=1 corresponds to 100Hz frequency. Hence, the resolution in frequency is increased. Since N ≥ L, increasing N to 8 from 80 for the same x[n] implies x[n] is still the same sequence (=8), the rest of x[n] is padded with zeros. This implies that there is no further information in time domain, but the resulting spectrum has higher frequency resolution. This spectrum is known as "high density spectrum" (resulting from zero padding x[n]). Instead of zero padding, for higher N, if more number of points of x[n] are taken (more data in time domain), then the resulting spectrum is called a "high resolution spectrum".

PROCEDURE

1. Open MATLAB
2. Open new M-file
3. Type the program
4. Save in current directory
5. Compile and Run the program
6. For the output see command window/Figure window

THEORETICAL CALCULATIONS

manual calculation of IDFT

$X(K) = \{4, 3.9959 - j0.1096, -0.6180 - j0.00003, -1.4239 + j0.5837, 1.9270 + j0.9511\}$

$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(K) e^{j\left(\frac{2\pi}{N}\right)kn}$.

$x(n) = \frac{1}{5} \sum_{k=0}^{4} X(K) e^{j\left(\frac{2\pi}{N}\right)kn}$.

for n=0

$x(0) = X(0) e^{j\left(\frac{2\pi}{5}\right)0} + X(1) e^{j\left(\frac{2\pi}{5}\right)0} + X(2) e^{j\left(\frac{2\pi}{5}\right)0} + X(3) e^{j\left(\frac{2\pi}{5}\right)0} + X(4) e^{j\left(\frac{2\pi}{5}\right)0}$

$x(0) = \frac{1}{5}[5] = 1$

for n=1

$x(1) = X(0) e^{j\left(\frac{2\pi}{5}\right)0} + X(1) e^{j\left(\frac{2\pi}{5}\right)1} + X(2) e^{j\left(\frac{2\pi}{5}\right)2} + X(3) e^{j\left(\frac{2\pi}{5}\right)3} + X(4) e^{j\left(\frac{2\pi}{5}\right)4}$

$x(1) = 2.$

for n=2

$x(2) = X(0) e^{j\left(\frac{2\pi}{5}\right)0} + X(1) e^{j\left(\frac{2\pi}{5}\right)2} + X(2) e^{j\left(\frac{2\pi}{5}\right)4} + X(3) e^{j\left(\frac{2\pi}{5}\right)6} + X(4) e^{j\left(\frac{2\pi}{5}\right)8}$

$x(2) = 0.$

for n=3

$x(3) = X(0) e^{j\left(\frac{2\pi}{5}\right)0} + X(1) e^{j\left(\frac{2\pi}{5}\right)3} + X(2) e^{j\left(\frac{2\pi}{5}\right)6} + X(3) e^{j\left(\frac{2\pi}{5}\right)9} + X(4) e^{j\left(\frac{2\pi}{5}\right)12}$

$x(3) = 0$

for n=4

$x(4) = X(0) e^{j\left(\frac{2\pi}{5}\right)0} + X(1) e^{j\left(\frac{2\pi}{5}\right)4} + X(2) e^{j\left(\frac{2\pi}{5}\right)8} + X(3) e^{j\left(\frac{2\pi}{5}\right)12} + X(4) e^{j\left(\frac{2\pi}{5}\right)16}$

$x(4) = 1$

output $x(n) = \{1, 2, 0, 0, 1\}$.

MATLAB PROGRAM TO FIND DFT

```
M = abs(XK);
M_db = 20*log(abs(XK));
a_deg = angle(XK)*180/pi;
Subplot(2,2,3);
Plot(k, m_db);
Xlabel('freq index k');
ylabel('magnitude (dB)');
title('magnitude spectrum');
Subplot(2,2,4);
plot(k, a_deg);
xlabel('freq index k');
ylabel('phase (degrees)');
title('phase spectrum');
```

Mathematical Calculations :-

$x(n) = \{1, 2, 0, 0, 1\}$

the given sequence is $x(n) = \{1, 2, 0, 0, 1\}$

no. of element in the sequence $N = 5$

The $DFT[x(n)] = X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\left(\frac{2\pi}{N}\right)nk}$

where $k = 0, 1, 2 \ldots N-1$

At $k = 0$

$$X(k) = \sum_{n=0}^{5-1} x(n)\, e^{-j\left(\frac{2\pi}{5}\right)nk}$$

$$X(0) = \sum_{n=0}^{4} x(n)$$

$$X(0) = [x(0) + x(1) + x(2) + x(3)] + x(4)]$$

$$X(0) = 4.$$

At $k = 1$

$$X(1) = \sum_{n=0}^{4} x(n)\, e^{-j\left(\frac{2\pi}{5}\right)1 \cdot n}$$

$$X(1) = x(0)\, e^{-j\left(\frac{2\pi}{5}\right)0} + x(1)\, e^{-j\left(\frac{2\pi}{5}\right)} + x(2)\, e^{-j\left(\frac{2\pi}{5}\right)2} + x(3)\, e^{-j\left(\frac{6\pi}{5}\right)} + x(4)\, e^{-j\left(\frac{8\pi}{5}\right)}$$

$$X(1) = 1 + 2\left[\cos\left(\frac{2\pi}{5}\right) - j\sin\left(\frac{2\pi}{5}\right)\right] + 1\left[\cos\left(\frac{8\pi}{5}\right) - j\sin\left(\frac{8\pi}{5}\right)\right]$$

$$X(1) = 1 + 2\left[0.99976 - j\,0.02194\right] + 1\left[0.99616 - j\,0.8762\right]$$

$$X(1) = 3.99568 - j\,0.10956.$$

$$X(2) = \sum_{n=0}^{4} x(n)\, e^{-j\left(\frac{2\pi}{5}\right)2 \cdot n}$$

$$= x(0) + x(1)\, e^{-j\left(\frac{4\pi}{5}\right)} + x(2)\, e^{-j\left(\frac{2\pi}{5}\right)4} + x(3)\, e^{-j\left(\frac{2\pi}{5}\right)6} + x(4)\, e^{-j\left(\frac{2\pi}{5}\right)\cdot 8}$$

$$= -0.618033 - j\,0.000314$$

for $n = 2$

$x(2) = x(0)$

$x(3)$

$x(2) = 4$

for $n = 3$

$x(3) = x(0)$

$+ x(3)$

$x(3) = 0$

for $n = 4$

$x(4) = x$

$+ x(3)$

$x(4) = 1$

output

ATLAB PROGRAM T

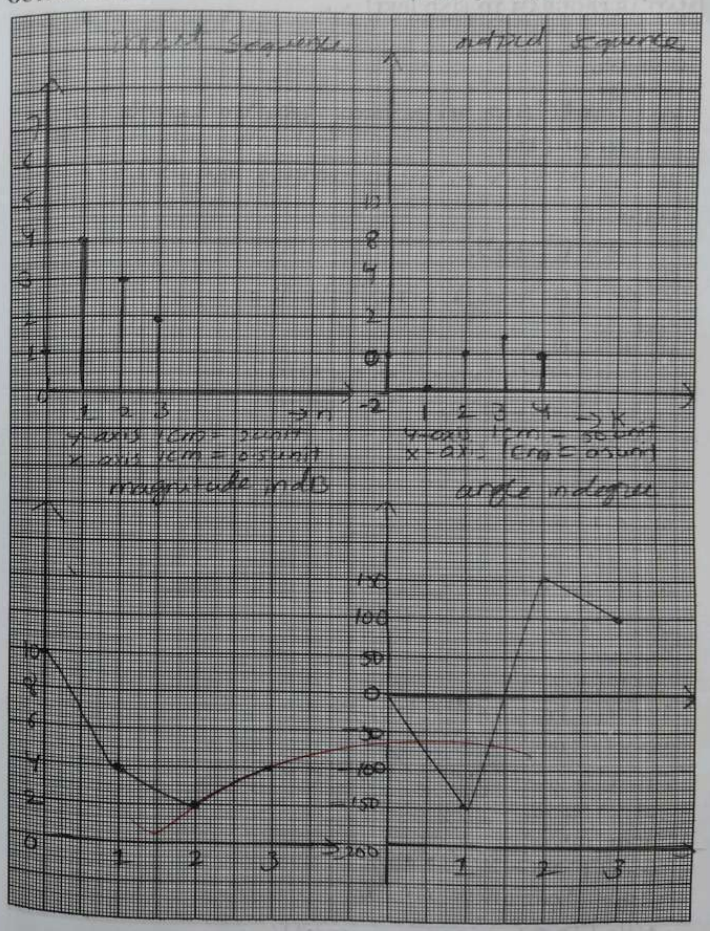
```
M = abs(X
M_db = 20*
a_deg = an
Subplot (2,2,3)
Plot (k, m_db)
Xlabel ('freq
ylabel ('magnit
title ('magnitu
subplot (2, 2,4);
Plot (k, a_deg
clabel ('freq
ylabel ('phase(
itle ('phase spe
```

```
clc;
Clear all; -> clear the variable all.
Close all; -> close figure all
x = input ('Enter the time domain sequence :');
L = length(x);
N = input ('Enter the output sequence length;');
if (N<L)
    disp('Error');
else
    x = [x zeros(1,N-L)];
end
for n=0: N-1
    for k= 0:N-1
        p = exp(-1) * 2 *pi * n*k * 1/N);
        W(n+1, k+1) = p;
    end
end
Xk = x*W;
n=0:1:N-1;
k=0:1:N-1;
disp ('DFT sequence');
disp('Xk');
subplot(221);
stem(n,x);
xlabel ('Time index n');
ylabel ('Amplitude x(n)');
title ('Time domain sequence');
subplot (2,2,2);
stem (k, Xk);
Xlabel ('freq index k');
Ylabel ('Amplitude X(k)');
title ('freq domain sequence');
```

[∵ abs → only +ve
remove -ve]

x - axis 1cm = 1unit
y - axis 1cm = 1unit
y - axis 1cm = 2 unit

OUTPUT: DFT

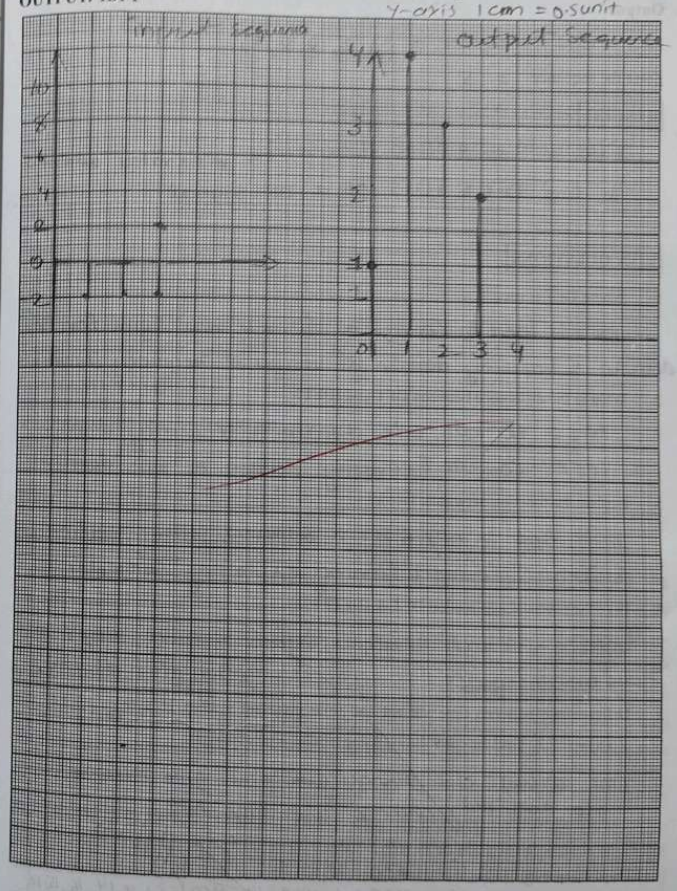


MATLAB PROGRAM TO FIND IDFT

```
clc;
clear all;
close all;
XK = input('Enter the frequency domain sequence : ');
L = length(XK);
N = input('Enter the output sequence length : ');
if(N<L)
    disp('Error');
else
   XK = [XK zeros[1, N-L]]; [∵ XK ∘
end
for n=0: N-1
      P = exp(1j*2*pi*n*k*1/N);
      W(n+1, k+1) = P;
    end
end
x = XK * W/N;
n = 0:1:N-1;
k = 0:1:N-1;
disp('IDFT sequence');
disp('xn');
subplot(2,1,1);
stem(k, XK);
xlabel('Freq index k');
ylabel('Amp X(k)');
title('frequency domain sequence');
subplot(2,1,2);
stem(n,x);
xlabel('Time Index n');
ylabel('Amplitude x(n)');
title('Time Domain sequence');
```


OUTPUT: IDFT

X-axis 1cm = 1unit
Y-axis 1cm = 2unit
Y-axis 1cm = 0.5unit



4. Rotate the inner circle one sample at a time in counterclockwise direction and go to step 3 to obtain the next value of output.
5. Repeat step No.4 until the inner circle first sample lines up with the first sample of the exterior circle once again.

- **Matrix Multiplication Method**

In this method, the circular convolution of two sequences x1(n)and x2(n)can be obtained by representing the sequences in matrix form as shown below

$$\begin{bmatrix} x_2(0) & x_2(N-1) & x_2(N-2) & \dots & x_2(2) & x_2(1) \\ x_2(1) & x_2(0) & x_2(N-1) & \dots & x_2(3) & x_2(2) \\ x_2(2) & x_2(1) & x_2(0) & \dots & x_2(4) & x_2(3) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_2(N-2) & x_2(N-3) & x_2(N-4) & \dots & x_2(0) & x_2(N-1) \\ x_2(N-1) & x_2(N-2) & x_2(N-3) & \dots & x_2(1) & x_2(0) \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_1(1) \\ x_1(2) \\ \vdots \\ x_2(N-2) \\ x_1(N-1) \end{bmatrix} = \begin{bmatrix} x_3(0) \\ x_3(1) \\ x_3(2) \\ \vdots \\ x_3(N-2) \\ x_3(N-1) \end{bmatrix}$$

The sequence x2(n) is repeated via circular shift of samples and represented in N * N matrix form. The sequence x1(n) is represented as column matrix. The multiplication of thesetwomatrices gives the sequences x3(n).

PROCEDURE

1. Open MATLAB
2. Open new M-file
3. Type the program
4. Save in current directory
5. Compile and Run the program
6. For the output see command window/Figure window

THEORETICAL CALCULATIONS: CONCENTRIC CIRCLE METHOD

$x(n) = \{2, 2, 1\}$ & $h(n) = \{2, 3, 1\}$

$$\begin{bmatrix} 2 & 1 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 11 \\ 10 \end{bmatrix}$$

$y(n) = \{9, 11, 10\}$

THEORETICAL CALCULATIONS: ~~MATRIX~~ Linear Convolution MULTIPLICATION METHOD

$x_1(n) = \{2, 2, 1\}$ $x_2(n) = \{2, 3, 1\}$

The given sequences are $x(n) = \{2, 2, 1\}$ & $h(n) = \{2, 3, 1\}$

length of $h(n)$ is $m = 3$

Length of $x(n)$ is $L = 3$

length of $y(n) = N = L + m - 1$

$N = 5$

	2	2	1
2	4	4	2
3	6	6	3
1	2	2	1

$y(n) = \{4, 10, 10, 5, 1\}$

MATLAB PROGRAM: CONCENTRIC CIRCLE METHOD

```
clc;
close all;
clear all;
x(n) = input ("enter first sequence");
h(n) = input ("enter second sequence");
  L = length (x(n));
  m = length (m(n));
  N = max (L,m);
x(n) = [ x(n) zeros (1, N-L)];
h(n) = [h(n) zeros(1,N-m)];
for n = 0:N-1;
    y(n+1)=0;
    for i: 0':N-1;
```



```
    j = mod(n-i,N);
    y(n+1) = y(n+1) + x(i+1)*h(j+1);
  end
end
disp ('output sequence');
disp (y);
subplot (311);
stem(x);
xlabel('time');
ylabel('amplitude');
title ('input sequence x');
subplot (312);
stem(h);
xlabel('time');
ylabel('amplitude');
title('Input sequence h');
subplot(313);
stem(y);
xlabel('time');
ylabel('amplitude');
title('circular convolu-
  -tion output y');
```

o/p:-
enter the first sequence: [2, 2, 1]
enter the second sequence: [2, 3, 1]
output sequence: 9 11 [illegible]

MATLAB PROGRAM: ~~MATRIX~~ MULTIPLICATION ~~METHOD~~

linear convolution.

```
clc;
clear all;
close all;
x = input ('enter the first sequence:');
h = input ('enter the second sequence:');
L = length (x);
m = length (n);
N = L+m-1;
x = [x zeros (1, N-L)];
h = [h zeros (1, N-m)];
for n = 0:N-1;
    y(n+1) = 0;
    for i=0:N-1
```

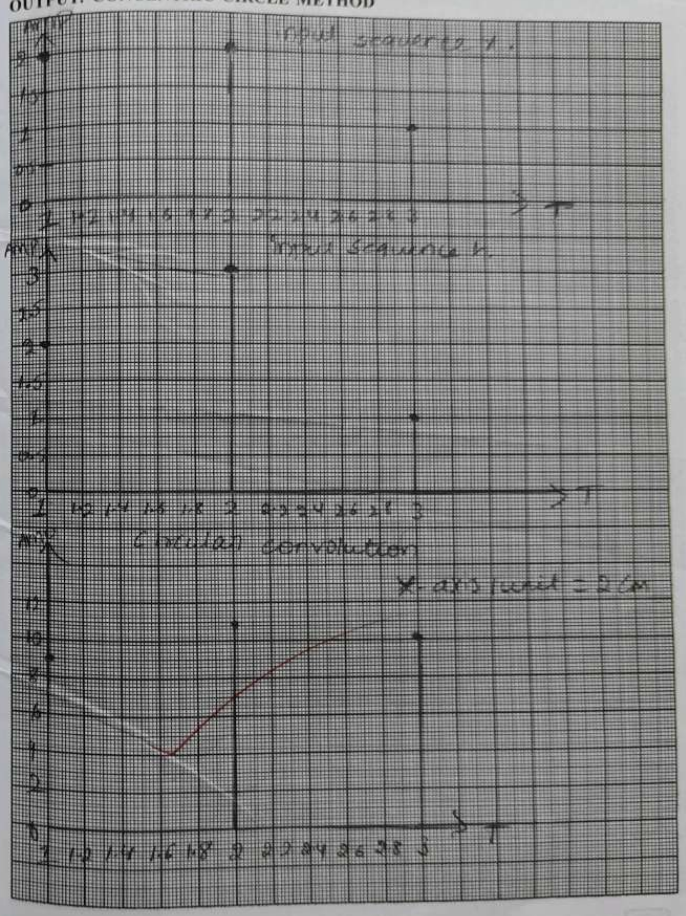
```
    if (n-i >= 0)
      j = mod(n-i,N);
      y(n+1)= y(n+1)+x(i+1)* h(n-i+1);
    end
end
disp('output Sequence:');
disp(y);
Subplot(311);
xlabel('time');
ylabel('Amplitude');
title('input Sequence x');
subplot(312);
stem(h);
xlabel('time');
ylabel('Amp');
title('input sequence h');
Subplot(313);
stem(y);
xlabel('time');
ylabel('Amp');
title('linear convolution output y');
```

o/p:-

enter the first sequences: [2, 2, 1]
enter the second sequences: [2,3,1]
output sequence
4 10 10 5 1.

X-axis 1unit = 0.2cm
Y-axis 1unit = 0.5cm

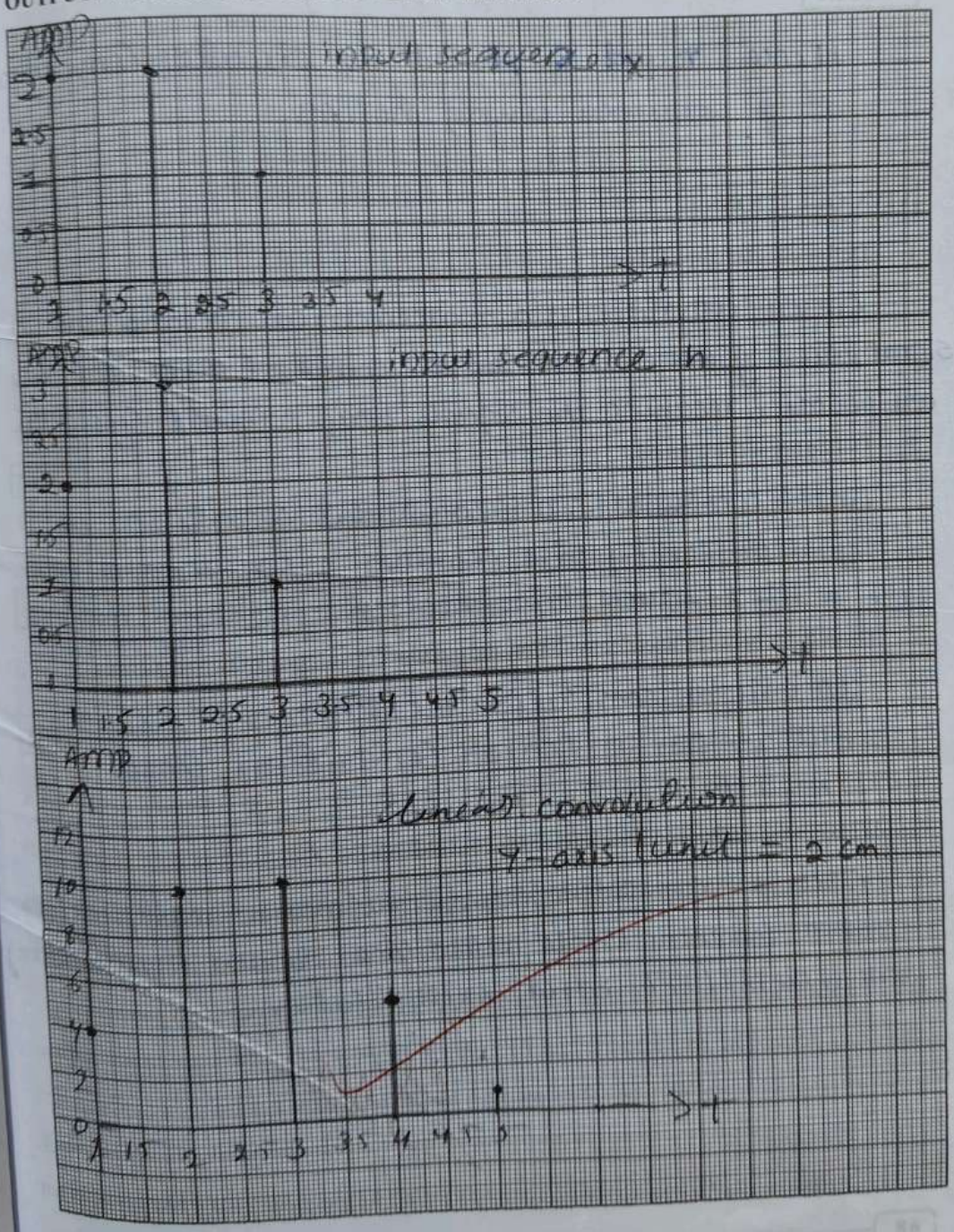
OUTPUT: CONCENTRIC CIRCLE METHOD



x - axis 1unit = 0.5cm
y-axis 1unit = 0.5cm

linear convolution

OUTPUT: ~~MATRIX~~ MULTIPLICATION METHOD



Theoritical Calculations

sequence is $x(n) = \{1, 2, 0, 0, 1, 0, 0\}$

length of sequence $L = 8$

Twiddle factors are

$$W_N^k = e^{-j\frac{2\pi}{N}\cdot k}$$

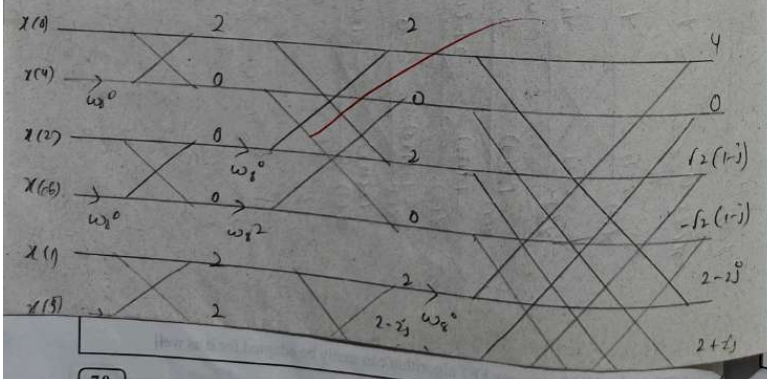
$$k = 0, 1, 2 \ldots N-1$$

$$W_8^0 = e^{-j\frac{2\pi}{8}\cdot 0} = 1$$

$$W_8^1 = e^{-j\frac{2\pi}{8}\cdot 1} = \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$$

$$W_8^2 = e^{-j\frac{2\pi}{8}\cdot 2} = -j$$

$$W_8^3 = -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} = -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$$



MATLAB PROGRAM

```
clc;
clear all;
close all;
x = input ('Enter the time domain sequence:');
L = length (x);
N = input ('Enter the output sequence length:');
if (N<L)
        disp('Error');
else
      x = [x zeros (1, N-L)];
end
Xk = fft (x, N);
n = 0:1 : N-1;
k = 0:1: N-1;
disp ('FFT sequence');
```

```
disp('XK');
subplot (2,2,1);
stem (n,X);
```

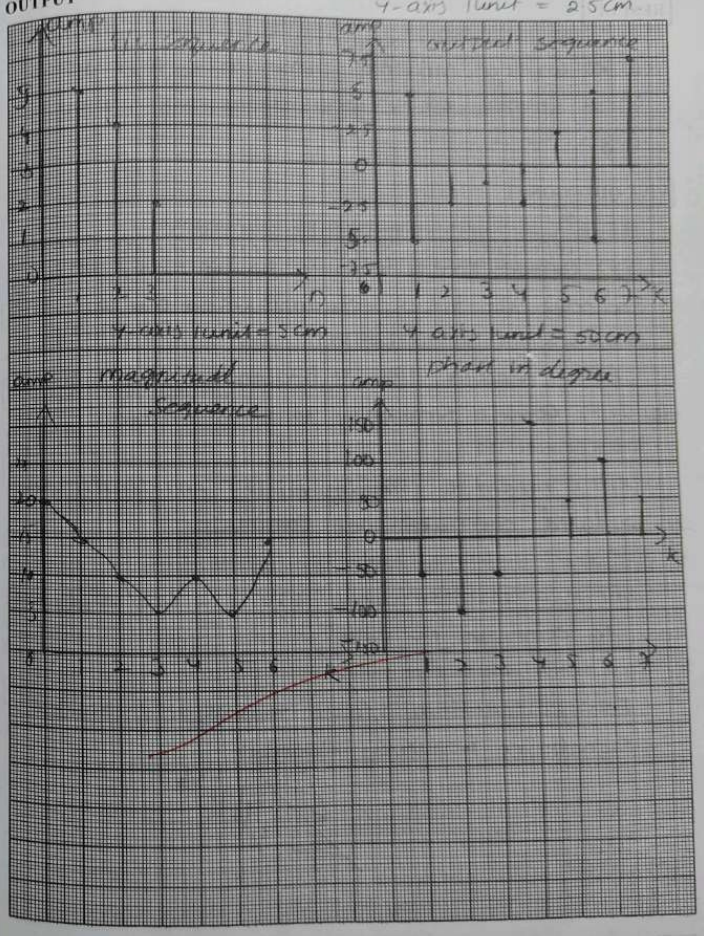
i/p:-

Sequence is $x(n) = \{1, 2, 0, 0, 1, 0, 0, 0\}$

length of sequence $L = 8$

x-axis 1unit = 1cm
y-axis 1unit = 1cm
y-axis 1unit = 2.5cm

OUTPUT



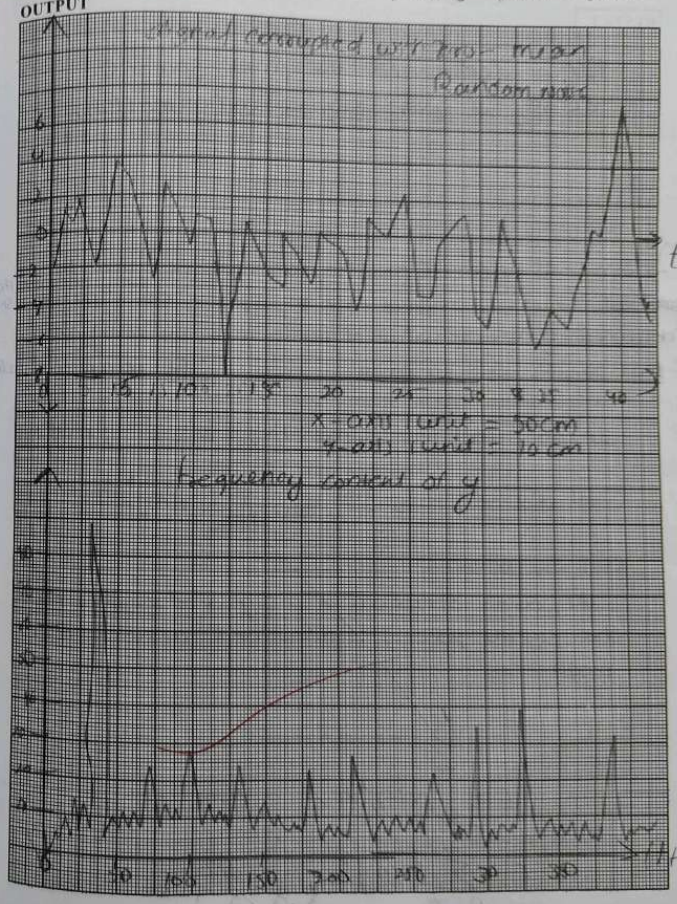
THEORETICAL CALCULATIONS

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt \quad \because \text{Cont}$$

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 \quad \because \text{dis}$$

MATLAB PROGRAM

```
clc;
clear all;
close all;
t = 0:0.001:0.6;
x = sin (2*pi * 50 * t);
y = x+2 * randn(size(t));
figure
plot (1000 * t (1:50), y(1:50))
title ('Signal corrupted with zero mean Ran-
    -dom Noise
xlabel ('time (milliseconds)');
Y = fft (Y, 512);
Pyy = Y.* conj(Y)/512;
f = 1000 * (0:256)/512;
figure
plot (f, Pyy (1:257));
title('freq content of y');
xlabel('freq (Hz)');
```


x-axis 1 unit = 2.5 cm
y-axis 1 unit = 2 cm.

OUTPUT